People simulation: wrong turns 7.4 % → 1.2 %

Nov 8, 2026 | Kamal | about 90 seconds

One-minute script

> In log 8 the rover turned one pillar too early. A person walking behind the rover looked, to the LiDAR, like the pillar we search for. We cannot tell people to stay away, so we built a simulation that adds people to the real recordings.
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> The method is on top of the slide. We take real LiDAR scans, add people as small circles, hide what is behind them, replay the same rules as the box, and compare the turn with the turn on the clean scan. We did this for 5,670 cases: people standing, walking along, towards and across, a follower, groups of two to five, and a crowd of eight.
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> The result is in the chart. With the old rule, 421 cases, 7.4 percent, turned at the wrong pillar. With version 10, which looks for the black sign structure beside the turn pillar, only 66, 1.2 percent. In 93.5 percent of cases the turn is within half a metre of the clean turn.
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> In the Open Sea the same idea: one person standing in the path made a false end wall in 18 of 20 runs; now in 0. A group of three, 20 to 0. A walking crowd, 13 to 0. Rain, snow and low sun cause no false stops. Still open: groups of three to five at the pillar before the sign, about 6 percent wrong turns, and this is simulation only.

What you need to understand

The idea: we cannot walk a crowd through every pillar, so we make the crowd in the computer and test our rules on it. This is a test of the rule, not a prediction of real crowds.

- The old rule looked for a pillar-like object. Every person standing near a pillar can look like one. The new rule needs the structure of the sign: it reaches from the pillar line to more than 2.3 m to the right, and it has to be seen several times from different positions.
- The turn also uses the place where the wall behind the pillars comes back after the gap (the 'B gap') and adds 5.35 m. This does not depend on where the run started.
- If nothing is found, the rover stops and asks instead of turning blindly.
- 'On target' means within ±0.5 m of where the clean scan turns. 'Wrong pillar' means more than 2 m too early. We show wrong turns in the chart because they are the dangerous ones; the old rule mostly turned late or not at all, which looks bad on 'on target' but is not as risky.
- Open Sea: 20 random starts per case, 12 for weather cases. The end wall must now be wide (≥ 1.1 m on both sides of the nearest surface), only after 85 percent of the crossing.

Likely questions

Are the people realistic? Circles of 0.15 to 0.30 m, speeds 0.8 to 1.6 m/s, placed 0.5 to 4 m from the track. Real people are less regular, so we also tested groups and a crowd.

Does the rover swerve around people? No. People in front: it stops and waits. It never steers around them.

What is still wrong? Groups of 3 to 5 at the pillar before the sign (about 6 percent wrong turns) and eight people shoulder to shoulder at the end wall.